

JAVA INTERFACE INHERITANCE PROGRAM

Application: Smart Home | Program 8

1. Aim

To write a Java program using interface inheritance where the SmartHome interface extends the Light and Fan interfaces, and the Home class implements SmartHome.

2. Steps for Execution

1. Open VS Code, Eclipse or IntelliJ IDEA with JDK installed.
2. Create a Java file named **Program8.java**.
3. Enter the Java source code and save the file.
4. Compile and run the program.
5. Create the Light interface with `switchOn()`.
6. Create the Fan interface with `switchFan()`.
7. Create SmartHome extending both Light and Fan, and add `showStatus()`.
8. Create the Home class implementing SmartHome.
9. Create a Home object and call `switchOn()`, `switchFan()`, and `showStatus()`.

3. Algorithm

1. Start the program.
2. Create the Light interface with `switchOn()`.
3. Create the Fan interface with `switchFan()`.
4. Create SmartHome extending Light and Fan.
5. Declare `showStatus()` in SmartHome.
6. Create Home implementing SmartHome.
7. Implement `switchOn()`, `switchFan()`, and `showStatus()`.
8. Create a Home object in `main()`.
9. Call all three methods.
10. Stop the program.

4. Explanation of the Program

Light interface: It declares the `switchOn()` method for controlling the light.

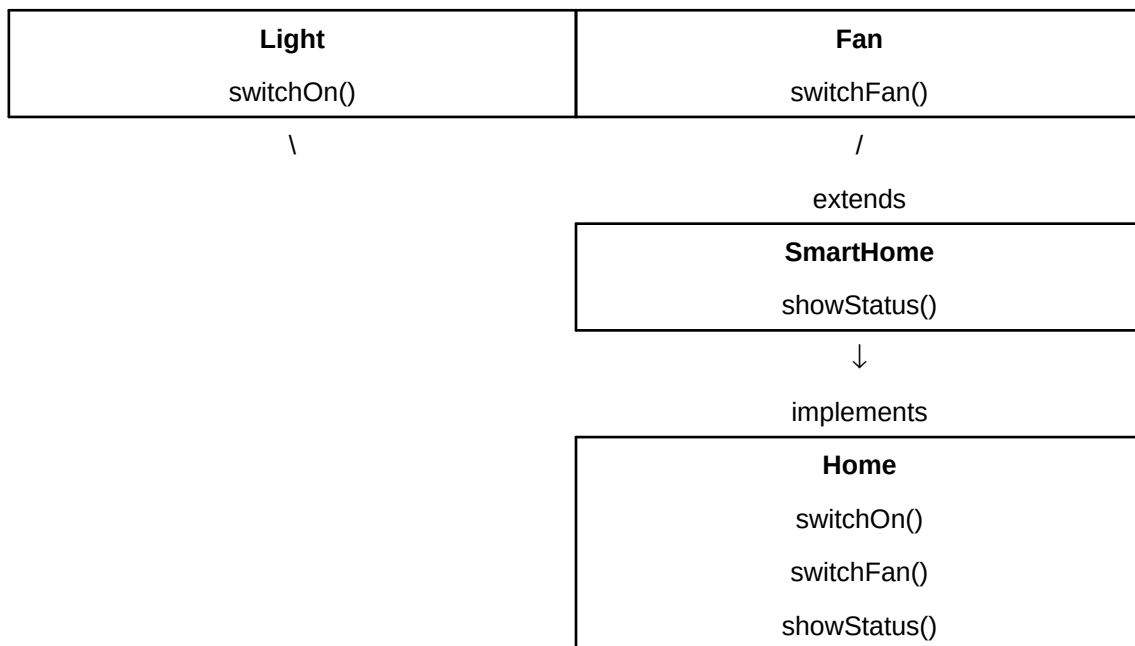
Fan interface: It declares the `switchFan()` method for controlling the fan.

SmartHome interface: It extends both `Light` and `Fan` and adds the `showStatus()` method.

Home class: It implements `SmartHome`, so it provides implementations for `switchOn()`, `switchFan()`, and `showStatus()`.

Interface inheritance: `SmartHome` inherits the methods of both `Light` and `Fan` through interface extension.

5. Interface Inheritance Diagram



6. Java Source Code

```
// Program 8: I1 extends I2, I3
// Application: Smart Home

interface Light {
    void switchOn();
}

interface Fan {
    void switchFan();
}

interface SmartHome extends Light, Fan {
    void showStatus();
}

class Home implements SmartHome {

    public void switchOn() {
        System.out.println("Light is ON");
    }
}
```

```
        public void switchFan() {
            System.out.println("Fan is ON");
        }

        public void showStatus() {
            System.out.println("Smart Home is active");
        }
    }

    public class Program8 {
        public static void main(String[] args) {

            Home h = new Home();

            h.switchOn();
            h.switchFan();
            h.showStatus();
        }
    }
```

7. Output of the Program

```
Light is ON  
Fan is ON  
Smart Home is active
```

8. Result

Thus, the Java program successfully demonstrates interface inheritance by extending the SmartHome interface from both Light and Fan and implementing it in the Home class.